Industrial Internship Report on

”Personal Expense Tracker”

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was Personal Expense Tracker This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 PREFACE

The internship provided me with an opportunity to apply my academic knowledge to practical software development. During this program, I worked on a project titled “Personal Expense Tracker,” which was developed using Java Swing, JDBC, and SQLite. The application helps users record and manage their income and expenses, maintain transaction records, calculate balances, and visualize spending patterns through charts.

An internship is important for career development because it provides practical experience, improves technical skills, and develops problem-solving abilities. Through this project, I gained knowledge of Java GUI development, database connectivity, transaction management, and application testing.

The main objective of the project was to provide a simple and efficient solution for managing personal finances. The internship opportunity provided by USC/UCT helped me apply my programming knowledge, understand the software development process, and improve my confidence in developing applications.

The work was carried out in stages, including requirement analysis, interface design, database creation, implementation, testing, and report preparation. This experience taught me the importance of systematic planning, debugging, and documentation.

I sincerely thank USC/UCT, Dhanalakshmi College of Engineering, the Department of Artificial Intelligence and Data Science, my faculty members, friends, and family for their support and encouragement throughout this learning experience.

2 Introduction

Personal financial management is an important part of everyday life. People regularly receive income and spend money on food, transportation, education, shopping, bills, and other personal requirements. Keeping track of these transactions helps individuals understand their spending habits, control unnecessary expenses, and plan their budgets effectively. However, recording transactions manually in notebooks or separate files can become difficult when the number of transactions increases. Calculating total income, total expenses, and remaining balances manually can also lead to errors.

The Personal Expense Tracker is a desktop application developed to simplify the process of recording and managing personal financial transactions. The application provides a graphical user interface through which users can enter income and expense details, view previously recorded transactions, edit or delete records, and search or filter transactions according to relevant criteria. It also provides financial summaries and charts that help users understand their spending patterns.

The application is developed using Java Swing for the graphical user interface, JDBC for database connectivity, and SQLite for local data storage. JFreeChart is used for presenting financial information graphically. The project demonstrates how programming concepts, database operations, and graphical components can be combined to develop a useful desktop application.

2.1 About Personal Expense Tracker

The Personal Expense Tracker is a Java-based desktop application developed to help users manage their personal financial transactions efficiently. The application provides a simple graphical interface for recording income and expenses, maintaining transaction records, and reviewing financial summaries. It aims to reduce the difficulties associated with manual expense tracking and make financial information easier to organize and understand.

The application is developed using Java Swing for the user interface, JDBC for database connectivity, and SQLite for storing transaction information locally. Users can add new transactions, edit or delete existing records, and search or filter transactions based on available criteria. The application also calculates total income, total expenses, and the remaining balance.

An important feature of the project is its ability to present financial information through charts. Category-wise expense charts help users identify spending patterns, while monthly income and expense comparisons provide a clearer overview of financial activity. These features make the application useful for individuals who want to maintain organized financial records.

The project demonstrates the practical application of Java programming, graphical user interface development, database management, and data visualization. It also provides experience in integrating different software components into a single desktop application.

2.2 Need and Scope of project

Managing personal finances requires regular monitoring of income and expenses. When transactions are recorded manually, users may experience difficulties calculating totals, finding previous records, correcting mistakes, and understanding their spending habits. A digital expense tracking application can simplify these activities by maintaining information in a structured format.

The Personal Expense Tracker addresses these needs by providing a centralized application for recording and managing financial transactions. SQLite stores the records locally, while Java Swing provides the interface through which users interact with the system. JDBC connects the application to the database and supports the required data operations.

The scope of the project includes transaction entry, record modification and deletion, searching and filtering, financial summary calculation, and graphical analysis of expenses. The application is intended for personal financial record management and provides a foundation for future improvements, such as budget planning, data export, recurring transaction support, and enhanced reporting.

Overall, the project demonstrates how a desktop application can make routine financial record keeping more organized, accessible, and convenient.

2.3 Objective

The main objective of the Personal Expense Tracker is to develop a user-friendly desktop application for managing personal income and expenses. The application helps users maintain financial records, calculate balances, and understand their spending patterns.

The specific objectives of the project are:

1. To develop a desktop application using Java Swing.
2. To provide functionality for adding, editing, and deleting income and expense transactions.
3. To store transaction details securely in a local SQLite database.
4. To implement database connectivity using JDBC.
5. To provide search and filtering options for locating transactions.
6. To calculate total income, total expenses, and the remaining balance.
7. To display category-wise expenses and monthly financial comparisons through charts.
8. To validate user inputs and reduce incorrect transaction entries.
9. To gain practical knowledge of Java programming, database management, and software development.

The overall objective is to make personal financial record keeping simpler, more organized, and convenient for users.

2.4 References

The development of the Personal Expense Tracker draws on technical concepts related to Java programming, GUI development, database connectivity, local data storage, and graphical visualization. The following official documentation resources are relevant to these technologies.

1. **Oracle Java Documentation**
   Provides information about Java language features, standard libraries, and application development.
   Website: https://docs.oracle.com/en/java/javase/21/

2. **Java Swing and Desktop API Documentation**
   Describes the Java desktop module and interface components used to create windows, forms, tables, buttons, and other GUI elements.
   Website: https://docs.oracle.com/en/java/javase/21/docs/api/java.desktop/module-summary.html

3. **Java Database Connectivity Documentation**
   Explains database access through JDBC, including connections, SQL statements, and result processing.
   Website: https://docs.oracle.com/en/java/javase/21/docs/api/java.sql/module-summary.html

4. **SQLite Documentation**
   Provides information about SQLite databases, SQL commands, table structures, and data storage.
   Website: https://www.sqlite.org/docs.html

5. **JFreeChart Documentation**
   Provides technical information about creating and displaying charts in Java applications.
   Website: https://javadoc.io/doc/org.jfree/jfreechart/latest/index.html

6. **Apache Maven Documentation**
   Explains project configuration, dependency management, and build processes for Java applications.
   Website: https://maven.apache.org/guides/

These resources provide technical background for the programming language, interface toolkit, database system, charting library, and build tool used in the project. The final report should list the resources that were actually consulted during development.

2.5 Existing System

The existing approach to personal expense management often relies on manual records, calculators, or spreadsheet applications. In a manual system, users write down their income and expenses and calculate totals separately. While this method can work for a limited number of transactions, it becomes less convenient when records accumulate over time.

A manual system may require users to review multiple entries to find a particular transaction. It also depends on users performing calculations correctly and updating totals whenever a record changes. If records are incomplete or disorganized, reviewing financial activity can become difficult.

Spreadsheet-based methods provide features such as sorting, formulas, and filtering. However, users may need to prepare the spreadsheet structure, maintain formulas, and configure charts to obtain meaningful summaries. The experience also depends on the user's familiarity with spreadsheet tools.

The common limitations of these approaches include:

- Time-consuming entry and review of financial records.
- Possibility of calculation and data-entry errors.
- Difficulty finding older transactions.
- Additional effort required to update records and totals.
- Limited visualization unless charts and formulas are configured.
- Risk of inconsistent record keeping when information is stored in different places.

The Personal Expense Tracker provides a dedicated interface for managing these activities in one application. However, this comparison describes general limitations of manual and spreadsheet-based approaches; it does not imply that every existing system has all these limitations.

2.6 Proposed System and Glossary

Proposed System

The proposed system is a desktop application that integrates transaction management, database storage, financial calculations, and graphical analysis. It is designed to provide the main functions required for maintaining personal income and expense records through a single interface.

The application uses Java Swing to display transaction forms, tables, summary information, and chart panels. When users enter or modify transaction details, the application processes their actions and performs the required database operations through JDBC.

SQLite stores the transaction records in a local database. The transaction table contains fields such as transaction ID, type, category, amount, date, and note. This structured approach allows records to be inserted, retrieved, updated, and deleted.

The main features of the proposed system include:

- Adding new income and expense transactions.
- Editing and deleting saved transactions.
- Searching and filtering transaction records.
- Calculating total income, expenses, and balance.
- Displaying category-wise expense charts.
- Comparing monthly income and expenses.
- Storing transaction records locally.

The application is intended to be straightforward to use and suitable for individuals who want to maintain organized financial records. Since it uses local storage, it does not inherently provide online synchronization or access across multiple devices. Such capabilities could be considered as future enhancements.

Glossary

Term	**Definition**
Personal Expense Tracker	A desktop application for recording and analyzing personal financial transactions.
Java	A programming language used to implement the application.
Java Swing	A Java toolkit used to create desktop graphical interfaces.
GUI	Graphical User Interface through which users interact with the application.
JDBC	Java Database Connectivity, used to access databases from Java.

SQLite	A lightweight database engine used for local data storage.
SQL	Structured Query Language used to manage and retrieve database records.
Database	An organized collection of information.
Transaction	A recorded financial activity, such as income or an expense.
Income	Money received by the user.
Expense	Money spent by the user.
Balance	The difference between total income and total expenses for the records being summarized.
Category	A label used to group transactions, such as food, transport, or shopping.
CRUD	Create, Read, Update, and Delete operations used to manage records.
Data Validation	Checking input values against requirements before processing or storing them.
Search	Finding records that match specified keywords or criteria.
Filtering	Displaying records that satisfy selected conditions.
Data Visualization	Presenting data through charts or other visual representations.
JFreeChart	A Java library used to create charts and graphs.
Maven	A tool used to manage Java project builds and dependencies.
IDE	Integrated Development Environment used to write, run, and debug code.
User Interface	The part of the application through which users enter information and view results.
Local Database	A database stored on the user's computer.
Testing	Checking whether an application behaves as expected under specified conditions.

3 Problem Statement

Managing personal finances is an important part of daily life, as individuals need to keep track of their income, expenses, savings, and spending habits. However, many people find it difficult to maintain accurate records of their financial transactions. Traditional methods, such as recording expenses in notebooks or maintaining spreadsheets, require manual effort and may lead to calculation errors, missing entries, and difficulties in organizing financial information. These methods also make it challenging to identify spending patterns and understand how money is being used over time.

The main problem addressed by the **Personal Expense Tracker** project is the lack of a simple, organized, and efficient desktop application for recording and monitoring personal financial transactions. Without a proper tracking system, users may lose track of their daily expenses, exceed their budgets, and find it difficult to compare their income with their spending. Searching through old records, updating incorrect entries, and calculating the total income, expenses, and remaining balance can also become time-consuming.

To overcome these challenges, the proposed system is designed as a Java-based desktop application that allows users to manage their financial transactions in one place. It provides options to add, edit, delete, search, and filter income and expense records. The application stores transaction details, including transaction type, category, amount, date, and notes, in a local SQLite database using JDBC. It also calculates total income, total expenses, and the available balance, helping users understand their financial position more easily.

Furthermore, the system provides monthly summaries and graphical representations of expenses by category and monthly income-versus-expense comparisons. These features help users identify spending trends and make better financial decisions. Therefore, the project aims to reduce manual record-keeping, improve the accuracy and accessibility of financial information, and provide a user-friendly solution for personal expense management.

4 Existing and Proposed solution

The existing system for personal expense management mainly involves maintaining records using notebooks, spreadsheets, or basic applications. These methods require manual data entry and calculations, which can lead to errors, missing records, and difficulties in tracking expenses over time. Finding specific transactions and comparing monthly income with expenses can also be time-consuming. Additionally, traditional methods provide limited graphical analysis of spending patterns.

The proposed Personal Expense Tracker is a Java-based desktop application designed to overcome these limitations. It allows users to add, edit, delete, search, and filter income and expense transactions through a user-friendly interface. The application uses Java Swing for the interface, JDBC for database connectivity, and SQLite for storing transaction records locally. It automatically calculates total income, total expenses, and the available balance. It also provides monthly summaries, category-wise expense charts, and monthly income-versus-expense comparisons. Thus, the proposed system simplifies financial record management, reduces manual calculations, and helps users understand their spending habits and make better financial decisions.

4.1 Code submission h
https://github.com/gopireddyharshateja123/upskillcampus/tree/main

4.2 Report submission
https://github.com/gopireddyharshateja123/upskillcampus/blob/main/PersonalExpenseTracker_HarshaTeja_USC_UCT.pdf

5 Proposed Design/ Model

The proposed system follows a modular design that separates the user interface, application logic, and database operations. This structure improves organization and makes the application easier to maintain.

5.1 High Level Diagram (if applicable)

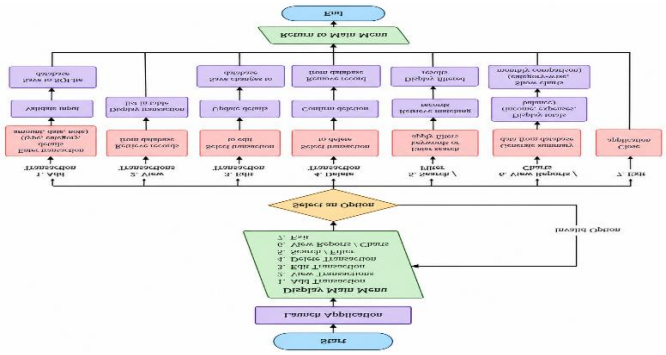
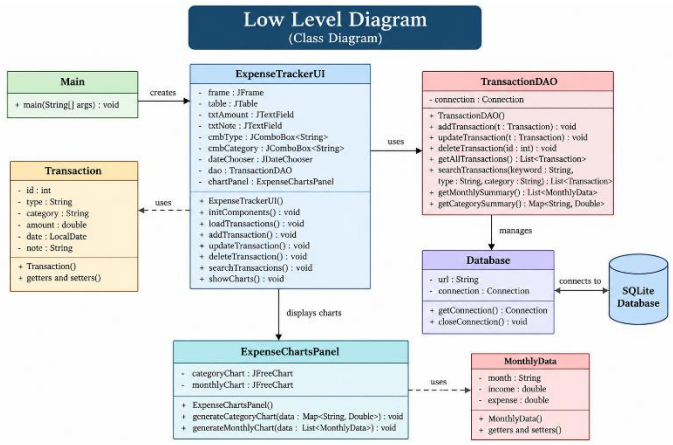


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The application provides a desktop interface for entering transaction details, viewing records in a table, and editing or deleting existing transactions. Search and filter options help users find specific records. Summary sections display total income, expenses, and balance, while charts present category-wise expenses and monthly income-versus-expense comparisons.

6 Performance Test

Performance testing is carried out to evaluate the functionality, reliability, and responsiveness of the Personal Expense Tracker application. The application is checked to ensure that transactions can be added, updated, deleted, searched, and displayed correctly. The database operations and financial calculations are also verified to ensure that the application produces accurate results.

6.1 Test Plan/ Test Cases

The test plan covers the major functions of the application, including transaction management, input validation, database operations, search and filtering, balance calculation, and graphical reports. Each feature is checked using suitable input data to verify whether the expected result is produced.

Test Case	Test Description	Expected Result
TC01	Add an income transaction	Transaction is saved successfully
TC02	Add an expense transaction	Expense record is stored
TC03	Edit an existing transaction	Updated details are displayed
TC04	Delete a transaction	Selected record is removed
TC05	Search by keyword or category	Matching records are displayed
TC06	Filter transactions by date range	Records within the selected range are displayed
TC07	Calculate income, expenses, and balance	Correct totals are displayed
TC08	View charts and monthly summary	Financial summaries and charts are displayed

6.2 Test Procedure

The application is launched, and each function is tested through its graphical user interface. Sample income and expense transactions are entered with the required details, such as type, category, amount, date, and notes. The records are then checked in the transaction table and database. Editing and deletion operations are performed to verify that changes are reflected correctly. Search and filtering options are

tested using different keywords, categories, and date ranges. Finally, the displayed financial totals, monthly summaries, and charts are checked against the available transaction data.

6.3 Performance Outcome

The application is designed to provide a simple and responsive interface for managing personal financial records. Its use of Java Swing supports desktop interaction, while SQLite enables local storage and retrieval of transaction data. Automated calculations and graphical summaries reduce the need for manual financial analysis. Functional testing should confirm that all operations work as expected. Actual execution times, measured performance results, and test pass/fail outcomes should be recorded after running the application; these results are not assumed in this report.

7 My learnings

During the development of the Personal Expense Tracker project, I gained practical knowledge of Java application development and learned how to build a desktop application with a graphical user interface. I improved my understanding of Java Swing components and learned how to design an interface that allows users to manage financial transactions easily.

I learned how to connect a Java application to an SQLite database using JDBC and perform database operations such as inserting, retrieving, updating, and deleting records. I also gained experience in organizing the application into separate classes, including the user interface, transaction model, database connection, and data access components.

Additionally, I learned how to implement search and filtering functions, calculate total income and expenses, and display financial information through charts and monthly summaries using JFreeChart. Testing different features helped me understand the importance of input validation, data accuracy, and error handling.

Overall, this project improved my programming, database management, problem-solving, and debugging skills. It also helped me understand the complete development process, from designing the application to implementing its features and verifying their functionality. The experience strengthened my confidence in developing practical software applications using Java and database technologies.

8 Future work scope

The Personal Expense Tracker can be further enhanced by introducing additional features to improve its usability, security, and financial management capabilities. In the future, the application can include a monthly budget planning feature that allows users to set spending limits for different categories and receive alerts when expenses approach or exceed those limits.

The application can also be enhanced with recurring transaction management, allowing users to record regular income and expenses automatically. Exporting financial reports in PDF, Excel, or CSV formats would help users maintain records and analyze their spending. Additional graphical dashboards could provide detailed insights into monthly and yearly financial trends.

Security can be improved by introducing user authentication and password protection to prevent unauthorized access to personal financial records. A backup and restore facility could help users protect their data and recover it when required. Furthermore, the application could be extended into a web or mobile application with secure cloud synchronization, allowing users to access their financial information across multiple devices.

These future enhancements would make the Personal Expense Tracker more flexible, secure, and useful for managing personal finances efficiently.